

CO1 Assessment Tool - 1 Test

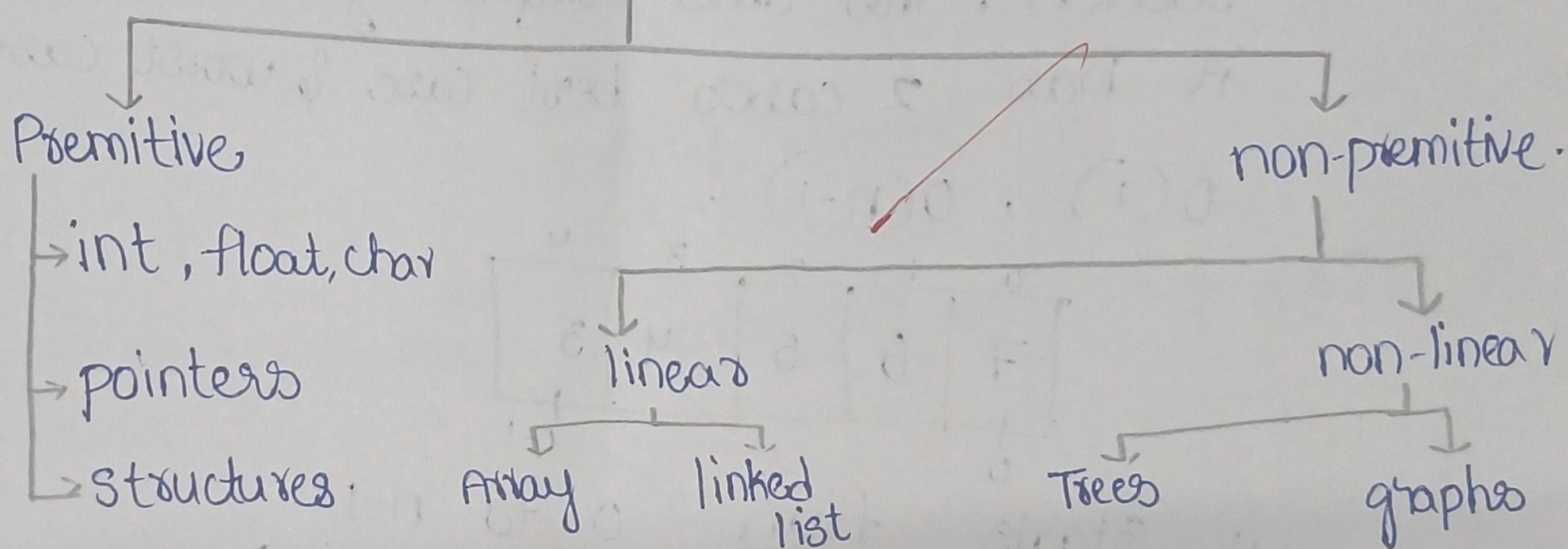
T. Deekshith

192425145
CSA0303.

22/30

i) Datastructure: It is the method of organizing the data and storing the data in the different ways and will organize the data to decrease the time.

datastructure.



- Array store vehicle data, speed, recovery, traffic coverage.
- linked list manage vehicle in queue at signals carry insertion & deletion.
- Array provide fast access.
- linked list allow easy inversion & deletion.

- Tree represents signal control hierarchy.
- Graph represents road network used for route planning.

2) linear search:

In a unsorted array, the searching of the element from the top to bottom in linear way is called linear search. The time complexity is Big-O notation. It has 2 cases best case & worst case is.

$O(1)$, $O(n-1)$.

0	1	2	3	4
7	6	5	4	3
0				n-1

It starts from 0 to n-1

```
for (int i = 0; i < n; i++) {
```

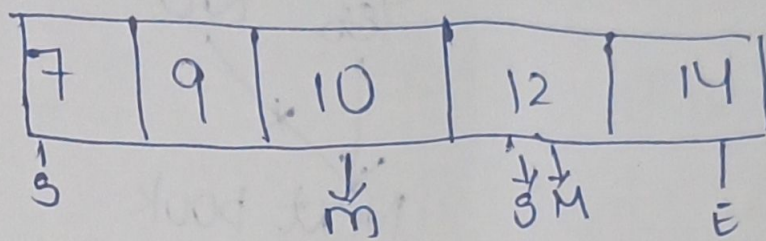
```
    int (arr[i] == key) {
```

```
        flag = 1; pos = i;
```

```
    }
```

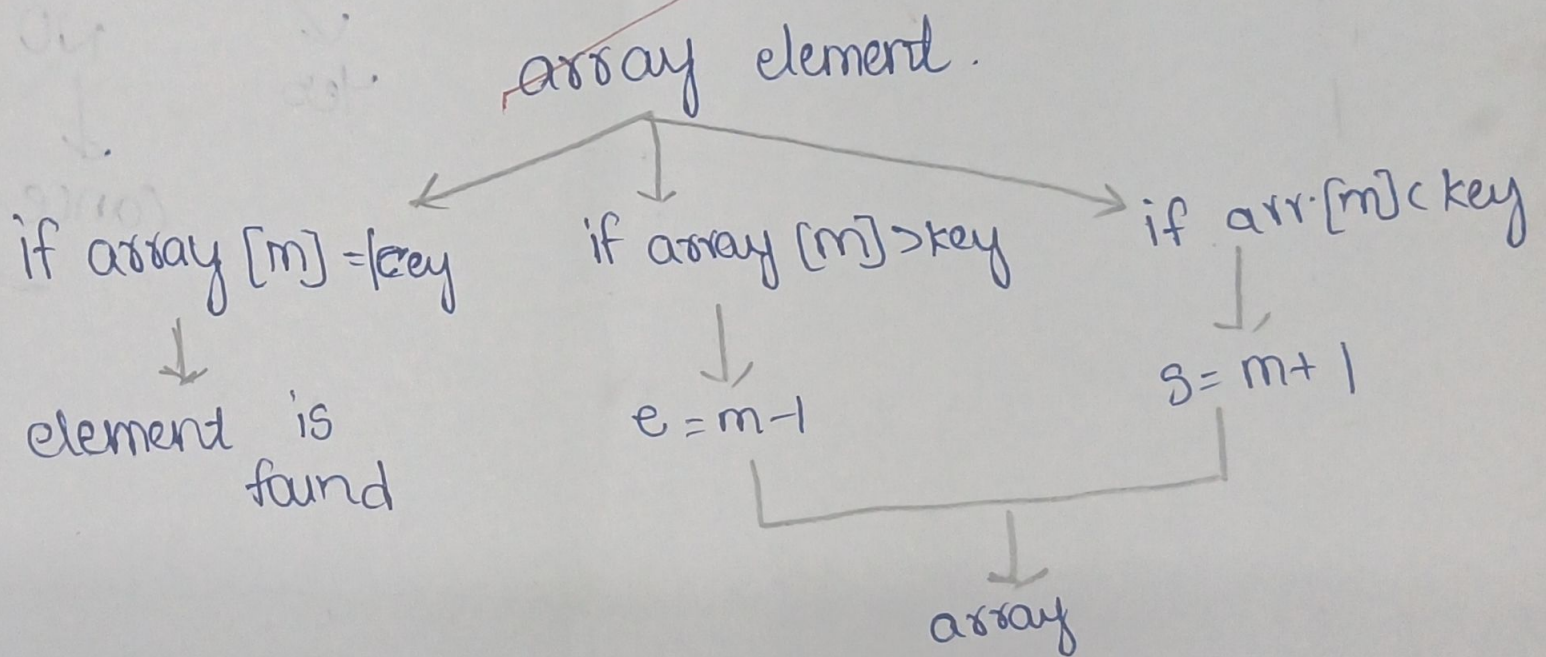
```
}
```


Binary search: The searching of element in a sorted array by using the divide & conquer method. The time complexity is $O(\log n)$, it has two cases best case & worst case is $O(\log n)$.



k = 12

- ① $m < k$, so $s = m + 1$
- ② $m == k$, key is found at 3.



→ This process will continue till $s < e$

3) library Management system .

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